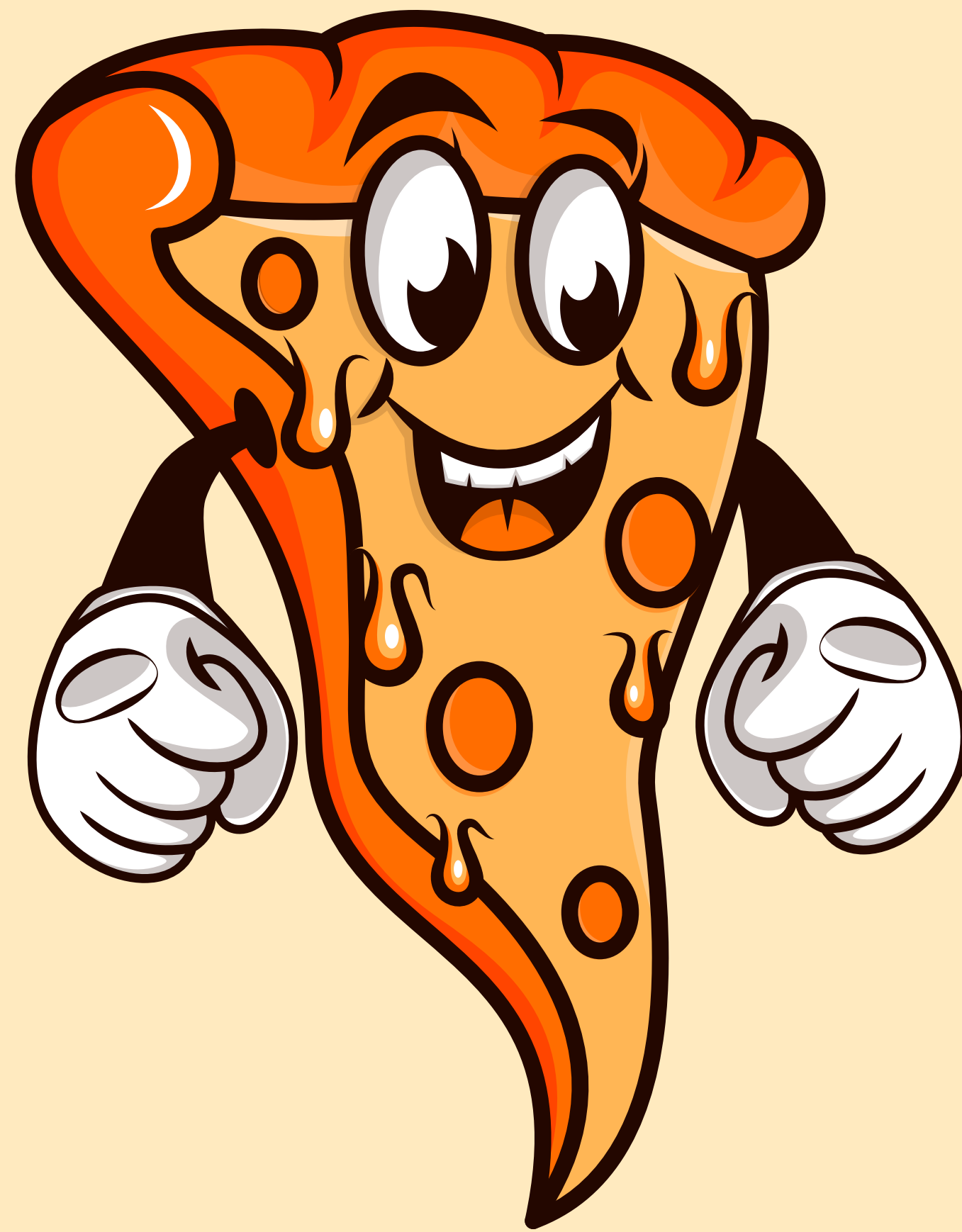




YUMMY PIZZA

BEST IN TOWN

 **Pizza Sales Analysis Using SQL**

**Transforming raw sales data into actionable
business insights using SQL.**

By Kaustubh Singh

Business Problem

Analyze pizza sales data to uncover revenue trends, customer preferences, and key business insights for data-driven decision-making. 🍕 📊

Tools Used

PostgreSQL – Database Management System

SQL – Data Querying & Analysis

pgAdmin 4 – Query Execution & Database Management

Canva – Presentation Design

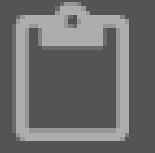
Q1/ RETRIVE THE TOTAL NUMBER OF ORDERS PLACED--

```
1/ RETRIVE THE TOTAL NUMBER OF ORDERS PLACED  
SELECT COUNT(order_id) AS Total_Order FROM orders
```

Data Output		Messages			
					
	total_order bigint 				
1	21350				

Q2/CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES.

```
SELECT SUM(os.quantity*p.price) AS total_revenue
FROM order_details AS os
JOIN pizzas p ON os.pizza_id=p.pizza_id
```

Data Output		Messages	
			
			
	total_revenue numeric		
1	817860.05		

Q3/IDENTIFY THE HIGHEST-PRICED PIZZA.

```
SELECT pt.name,  
       p.price  
FROM pizzas p  
JOIN pizza_types pt  
ON p.pizza_type_id = pt.pizza_type_id  
ORDER BY p.price DESC  
LIMIT 1;
```

Data Output

Messages

Notifications



SQL

name

character varying (100) 

price

numeric (5,2) 

1

The Greek Pizza

35.95

Q4/IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED

```
SELECT p.size,COUNT(p.size) FROM pizzas AS p
JOIN order_details od ON p.pizza_id=od.pizza_id
GROUP BY p.size
ORDER BY COUNT(p.size) DESC
```

Data Output		Messages	Notificatio				
							
	size character varying (5) 	count  bigint					
1	L	18526					
2	M	15385					
3	S	14137					
4	XL	544					
5	XXL	28					

Q5/LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES.

```
SELECT pt.name,SUM(od.quantity) FROM order_details AS od
JOIN pizzas p ON od.pizza_id=p.pizza_id
JOIN pizza_types pt ON p.pizza_type_id=pt.pizza_type_id
GROUP BY pt.name
ORDER BY SUM(od.quantity) DESC
LIMIT 5;
```

Data Output

Messages

Notifications

	name character varying (100)	sum bigint
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Piz...	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

Q6/JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA CATEGORY ORDERED.

```
SELECT pt.category,SUM(od.quantity)FROM pizza_types AS pt
JOIN pizzas p ON pt.pizza_type_id=p.pizza_type_id
JOIN order_details od ON p.pizza_id=od.pizza_id
GROUP BY pt.category
ORDER BY SUM(od.quantity) DESC
```

Data Output

Messages

Notifications

	category character varying (50)	sum bigint
1	Classic	14888
2	Supreme	11987
3	Veggie	11649
4	Chicken	11050

Q7/DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY.

```
SELECT  EXTRACT(HOUR FROM time) AS hour,
        COUNT(order_id) AS orders_per_hour
FROM    orders
GROUP BY EXTRACT(HOUR FROM time)
ORDER BY count(order_id) desc;
```

Data Output			Messages	Notifications	
			SQL		
	hour numeric	orders_per_hour bigint			
1	12	2520			
2	13	2455			
3	18	2399			
4	17	2336			
5	19	2009			
6	16	1920			

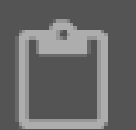
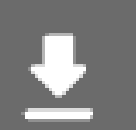
**Q8/GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE
NUMBER OF PIZZAS ORDERED PER DAY.**

```
SELECT ROUND(AVG(quantity),2)AS avg_pizza_per_day
FROM
(SELECT o.date,SUM(od.quantity) AS quantity FROM orders AS o
JOIN order_details od on o.order_id=od.order_id
GROUP BY o.date)
```

Data Output		Messages	
			
			
	avg_pizza_per_day 		
	numeric		
1	138.47		

Q9/DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE.

```
SELECT pt.name,round(SUM(od.quantity*p.price),0) AS revenue
FROM order_details AS od
JOIN pizzas p ON od.pizza_id=p.pizza_id
JOIN pizza_types pt ON p.pizza_type_id=pt.pizza_type_id
GROUP BY pt.name
ORDER BY revenue DESC
LIMIT 3;
```

Data Output			Messages	Notifications		
						
						
	name		revenue			
	character varying (100)		numeric			
1	The Thai Chicken Pizza		43434			
2	The Barbecue Chicken Piz...		42768			
3	The California Chicken Piz...		41410			

Q10/CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE.

```
SELECT pt.category,
ROUND(SUM(p.price*od.quantity)*100/
      ( SELECT SUM(p.price * od.quantity)
        FROM order_details od
        JOIN pizzas p
          ON od.pizza_id = p.pizza_id),2)
AS percent_contri
FROM order_details od
JOIN pizzas p
ON od.pizza_id = p.pizza_id
JOIN pizza_types pt
ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.category
ORDER BY percent_contri DESC;
```

	Data Output	Messages	Notifications
			
			
			
			SQL

	category character varying (50) 	percent_contri numeric 
1	Classic	26.91
2	Supreme	25.46
3	Chicken	23.96
4	Veggie	23.68

Q11/ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME.

```
SELECT date, revenue,
SUM(revenue) OVER (ORDER BY date) AS cumulative_sum
from
  (SELECT o.date, SUM(p.price*od.quantity) AS revenue
   FROM orders AS o
   JOIN order_details od ON o.order_id=od.order_id
   JOIN pizzas p ON od.pizza_id=p.pizza_id
   GROUP BY o.date
   ORDER BY o.date)
```

Data OutputMessagesNotifications

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SQL

	date 	revenue 	cumulative_sum 
1	2015-01-...	2713.85	2713.85
2	2015-01-...	2731.90	5445.75
3	2015-01-...	2662.40	8108.15
4	2015-01-...	1755.45	9863.60
5	2015-01-...	2065.95	11929.55

Q12/DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY.

```
SELECT * FROM
(
  SELECT pt.category,pt.name, SUM(p.price*od.quantity) AS revenue,
  RANK() OVER(partition by pt.category
  ORDER BY SUM(p.price * od.quantity)DESC) AS rn
  FROM order_details AS od
  JOIN pizzas p ON od.pizza_id=p.pizza_id
  JOIN pizza_types pt ON p.pizza_type_id=pt.pizza_type_id
  GROUP BY pt.category,pt.name
)
WHERE rn<=3
```

	category character varying (50) 	name character varying (100) 	revenue numeric 	rn bigint 
1	Chicken	The Thai Chicken Pizza	43434.25	1
2	Chicken	The Barbecue Chicken Piz...	42768.00	2
3	Chicken	The California Chicken Piz...	41409.50	3
4	Classic	The Classic Deluxe Pizza	38180.50	1
5	Classic	The Hawaiian Pizza	32273.25	2
6	Classic	The Pepperoni Pizza	30161.75	3
7	Supreme	The Spicy Italian Pizza	34831.25	1
8	Supreme	The Italian Supreme Pizza	33476.75	2